



Orb v1

Clean tyre-degradation curves from Friday, scored on Sunday.

Friday lies twice. We correct the lie you can measure and learn the one you can't.

Team Orb v1 · Samuel Christ
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Submitted to the idea round as ClearStint, renamed Orb v1. Built on disclosed pre-work from that round (estimator and 6-weekend validation, 5 Sep); everything else on these slides was built at Plaksha.





A Friday lap time is not a tyre measurement

lap time = fuel burn + track evolution + traffic + driver + push profile + tyre

Hungary 2026, medium tyre

+0.28 s/lap

a straight line through Friday's long runs

+0.04 s/lap

what the race showed — wrong by 7×

Two lies, two fixes

- Lie one, measurable: fuel, track evolution, traffic, driver and push profile are in the lap time. We strip them, lap by lap, from public telemetry.
- Lie two, not measurable on Friday: drivers attack tyres in practice and manage them on Sunday. We learn how much, per compound, from other weekends in leave-one-weekend-out evaluation, and score it every Sunday.
- When Friday has no signal, the curve is withheld. The fallback uses other withheld cases; its error and coverage remain visible.



What Orb v1 does, in six steps

Same estimator on practice and race, so the practice-to-race ratio is a property of the data, not of the method

1

Ingest

FastF1 timing archive: laps, 3.7 Hz telemetry, weather, race control. Telemetry quality checked lap by lap; degraded feeds refused.

2

Clean

Stint fixed effects absorb driver, car and fuel load. Fuel prior 1.1 kg/lap; track evolution measured from every driver's push laps; traffic laps dropped via gap-to-car-ahead.

3

Gate

A curve is issued only with 30+ clean long-run laps and a positive cleaned slope. Otherwise: withheld, with the reason.

4

Learn

Race $\div$ practice ratio per compound from other weekends in leave-one-weekend-out evaluation; applied when a majority agree within $\pm 50\%$.

5

Forecast

Curve $\times$ factor with a 90% band; withheld compounds get the fallback learned from other withheld cases.

6

Score

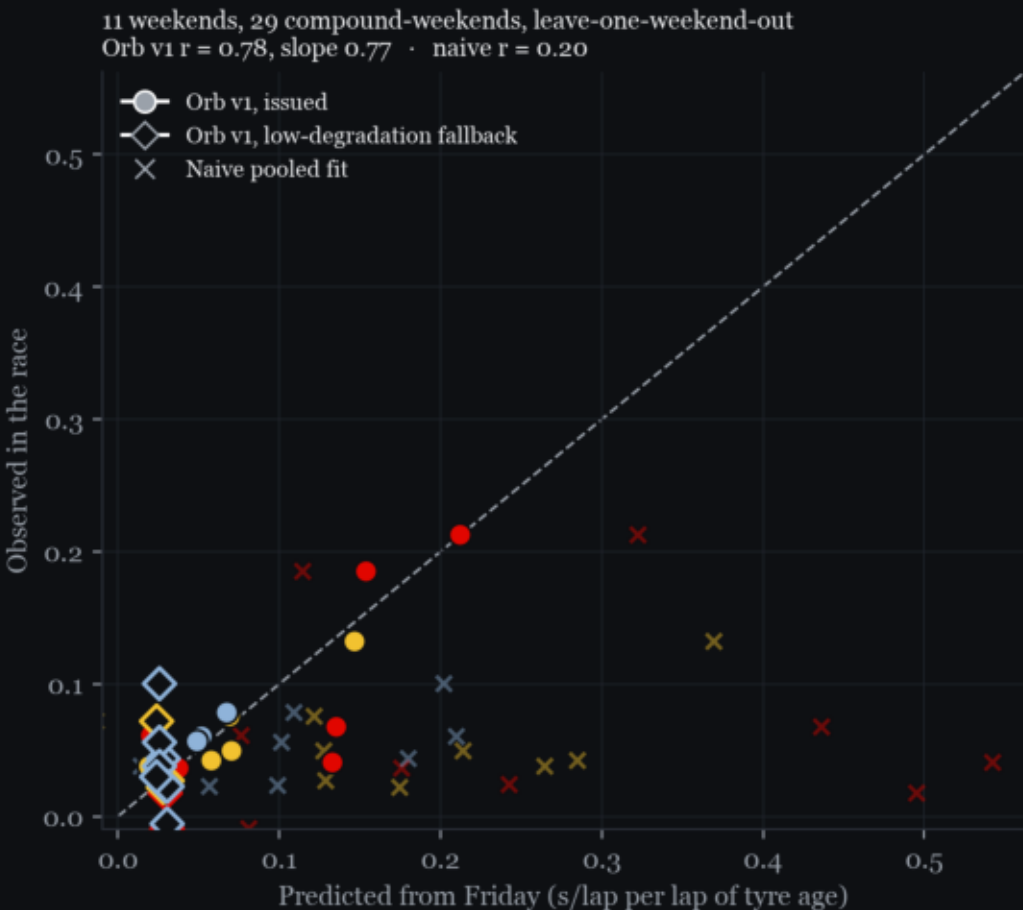
Sunday night: the race analysed with the same estimator; predicted vs observed, per compound, every weekend.

Friday transfers, once you learn how Sunday is managed

11 weekends, 29 compound-weekends, every number leave-one-weekend-out; the held-out weekend never sees its own race

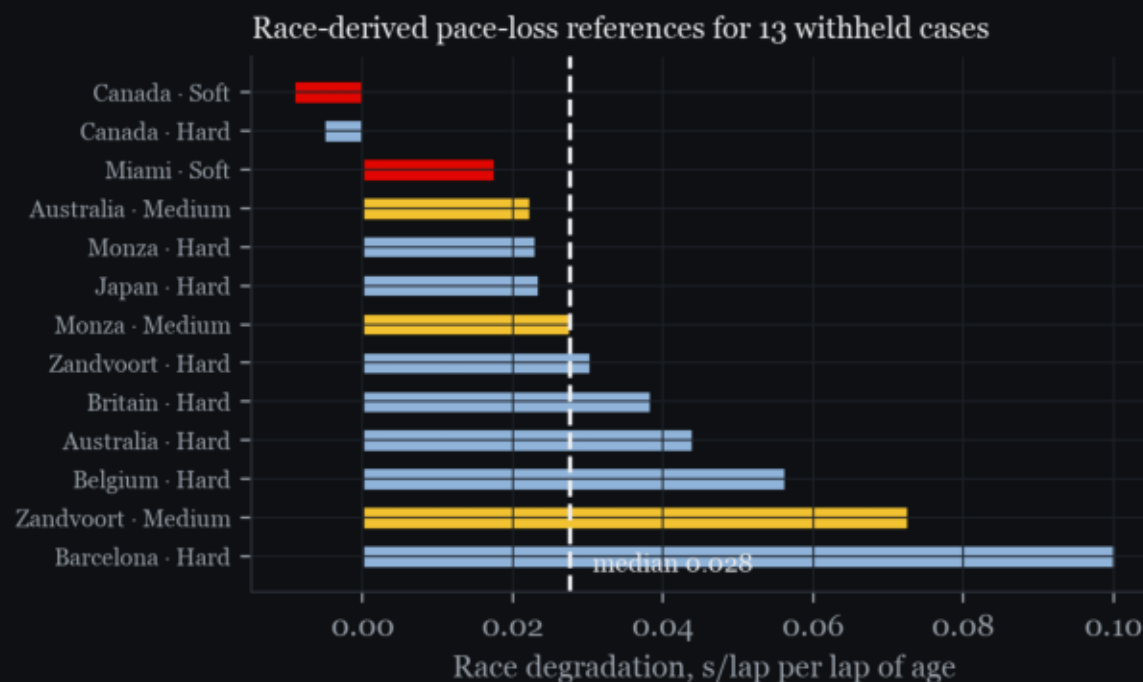
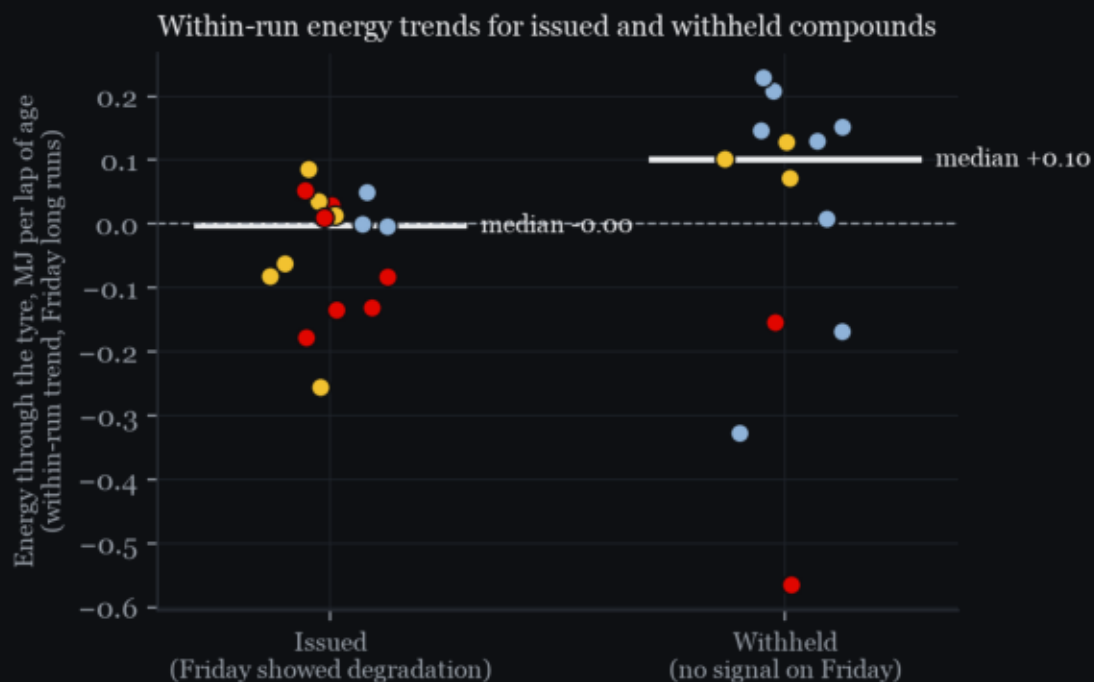
	MAE s/lap	calib. slope	r
Naive pooled fit	0.137	+0.07	+0.20
Cleaned Friday curve (issued)	0.045	+0.45	+0.67
Orb v1 (issued)	0.023	+0.76	+0.82
Orb v1, all cases incl. fallback	0.023	+0.77	+0.78

- Withheld compounds: median race-derived reference 0.028 s/lap; 11 of 13 below 0.06; band covered 9 of 13.
- Orb v1 beats the naive fit on 28 of 29 cases; 90% interval on its error 0.016 to 0.030 s/lap.
- Season transfer factors: medium $\times 0.45$ and soft $\times 0.97$; hard not applied. Each held-out weekend uses its own training pool.



The driver's push profile

Tyre energy per lap from position and speed traces; a public proxy

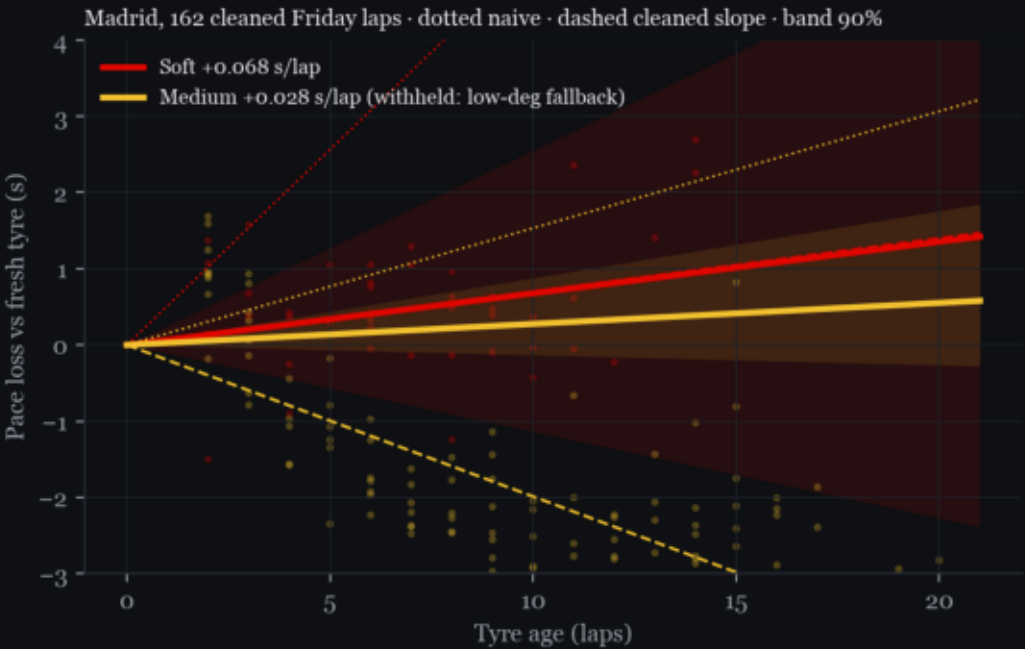


- Where the cleaned slope pointed the wrong way (9 cases), energy rose through the run in 9. 4 withheld cases had falling energy trends, so ramping is not universal.
- Practice and race energy coefficients are estimates, not physical constants. Their event-specific values are reported in the lock; the push-adjusted forecast is a second opinion.
- On 5 withheld cases with a push signal: push-adjusted MAE 0.029 vs fallback 0.028 s/lap. This subset differs from the full withheld pool.



Madrid: the frozen forecast

FP1, FP2, FP3, Q in the lock; published 21:33:14 IST. Hashed before the race, verifiable after the event.



Soft: +0.068 s/lap

90% band -0.114 to +0.255; n=46 clean practice laps

issued

Medium: +0.028 s/lap

90% band -0.013 to +0.088; n=112 clean practice laps

withheld: no positive degradation signal in cleaned practice

Central plan: S-M 22/34; band-high S-M-M. Pit loss 21 s. Hard has no issued curve.

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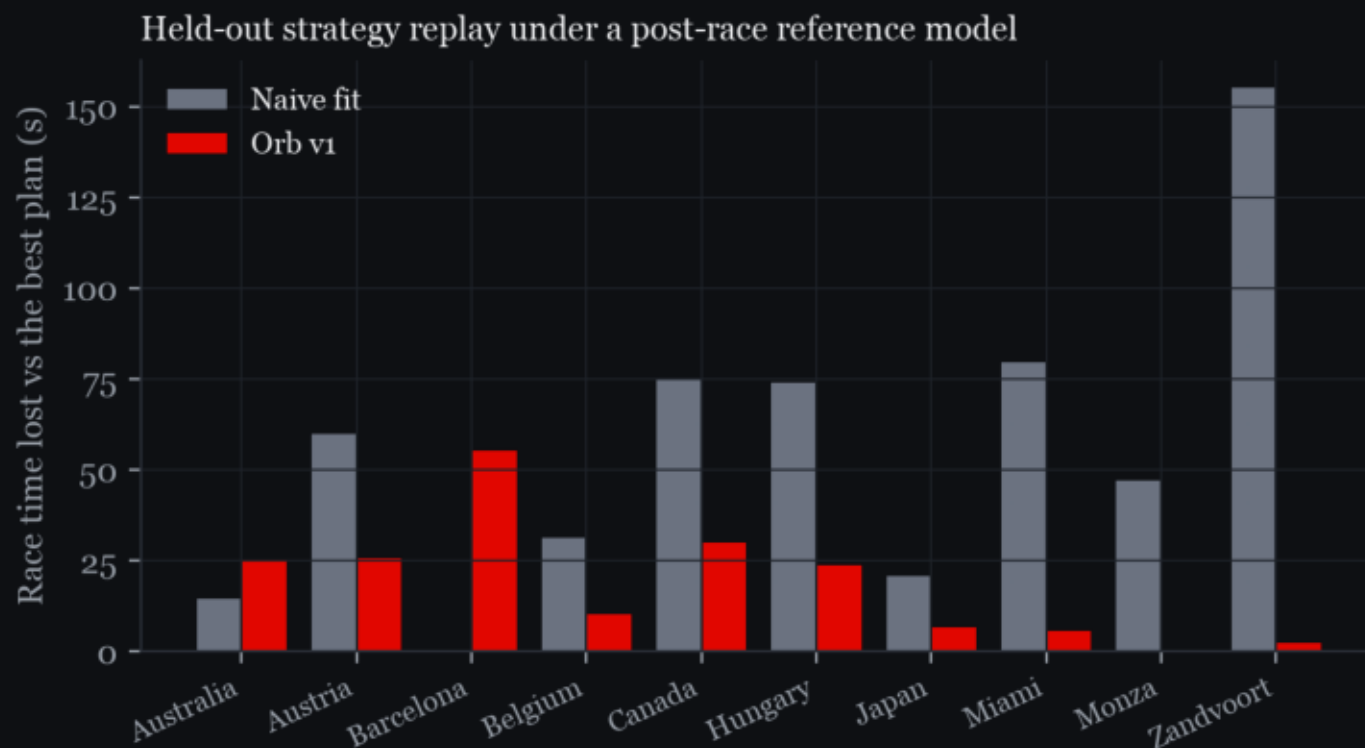
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SHA256 sidecar 30a5f06786fc332faf7934d0ed70cadd3bcba697f538939dfa5c8e905afe2c3c

lock v2 forecast_hash 66e3201e860b19efc410cac30073d6c12bd7d4f9cb894cd70a445747d3d0451a

The curve is only useful as a decision

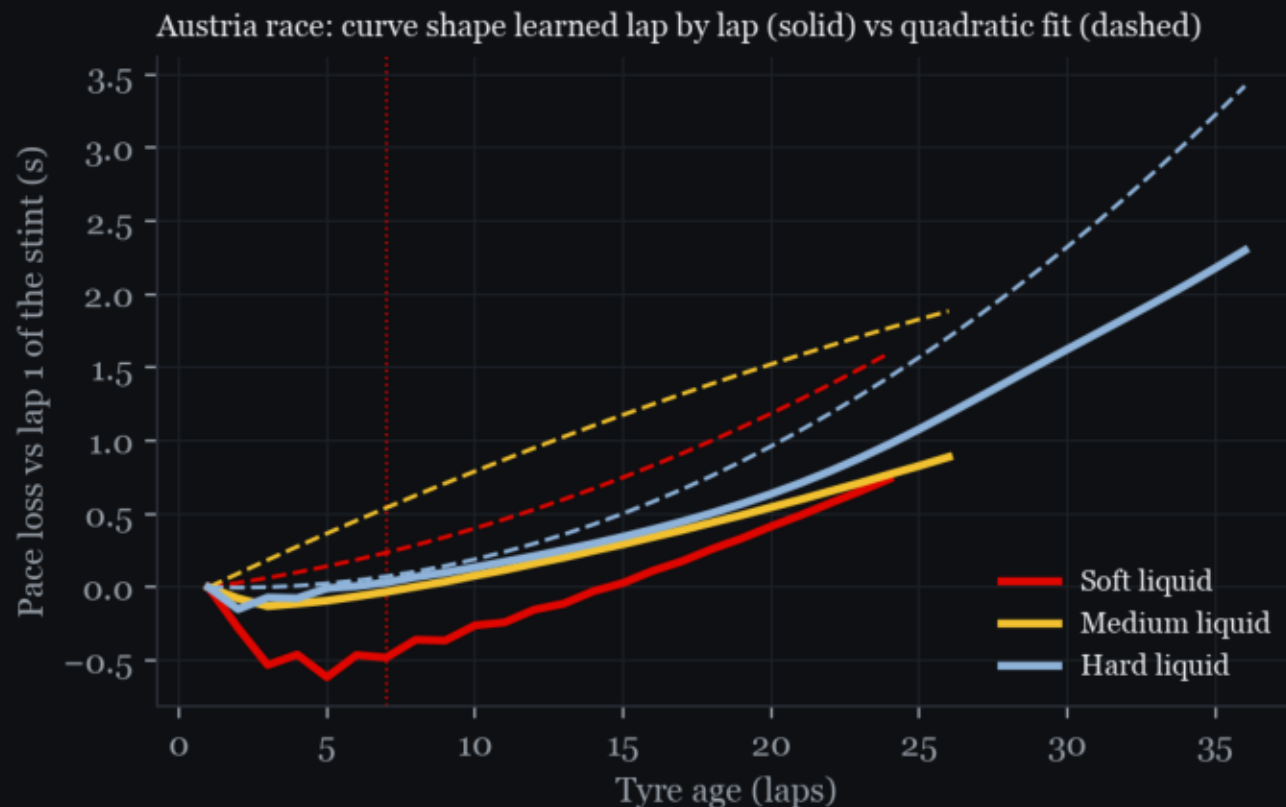
Held-out strategy replay under a post-race reference model; development pool, never sealed weekends



- Orb v1's plan beats the naive plan in 8 of 10 replays.
- Hungary: naive +74 s, Orb v1 +24 s. Monza: +47 s vs 0 s. Zandvoort: +155 s vs +2 s.
- The misses are Barcelona and Australia. The replay is model-implied; it does not claim observed race time saved.
- Offsets from practice or qualifying medians, with a nominal step for implausible measurements; 21 s pit loss; linear curves; no safety car or weather.

The neural model: an ablation, not the forecast

Race stints held out against linear, quadratic and covariate baselines



- The network beat the best baseline in 9 of 31 evaluated cells with the same per-lap inputs.
- This experiment tests whether a flexible model adds value to the inputs. It does not establish a tyre-wear sensor or a validated cliff detector.
- The deployed pre-race forecast remains the cleaned curve with its learned transfer and explicit fallback. The neural experiment remains an ablation.



The product: Live Predictor and Ghost Strategy

Every number traces to the lock, a hashed sidecar, a labelled computation or a placeholder

- Live weekend: issued and withheld cards, bands, the cleaned laps behind the curve, and why a compound was withheld.
- Sunday scorecard: predicted vs observed per compound, the week after.
- Strategy: crossover laps, one-stop vs two-stop, stint lengths, cost under the observed curves.
- Season validation: calibration, transfer factors, issued and withheld compounds.
- Excluded laps: every dropped practice lap with its reason. Method and assumptions: every prior, stated.
- Live Predictor updates a frozen prior from observed laps. Ghost Strategy explains model-implied alternative tyre plans.

Who this is for

- Categories with thin data and two engineers: F2, F3, F1 Academy, Indian F4, sim leagues, broadcasters. Public timing in, decision out, for any weekend by name.
- A works team has better data; the method transfers to it, the product serves the rest.
- Beyond the track, the same loop, measure the confounders, gate, learn the transfer, score against race-derived pace-loss references, fits fleet tyre wear and e-rickshaw battery health, where load, road, heat and driver stand in for fuel, evolution, traffic and push.



Built when, what it cannot do yet, and Sunday night

Disclosed pre-work vs Challenge Day build, as the rules ask

Disclosed pre-work (5 Sep)

- Feature extraction and the fixed-effects estimator.
- Six-weekend leave-one-out validation and the transfer factor rule.
- Hungary and Austria strategy replay.

Built at Plaksha

- 11 scored weekends incl. sprint format; Madrid live.
- The gate's low-degradation fallback and the push-profile diagnostic.
- Offsets with their measured or nominal basis; strategy replay.
- Liquid model and its ablation.
- Dashboard, lock file, one-command refresh.

Limits and next

- Linear curves; fuel prior assumed (a race-data estimate agrees at 0.029 s/kg).
- Headline error stays 0.021 to 0.027 s/lap across fuel prior 0.9–1.3 kg/lap, 0.025–0.035 s/kg, traffic 20–40%, runs 4–7 laps.
- Bands: raw 90% bands covered 76%; conformally widened leave-one-out they cover 86%.
- Separate development, sealed and prospective evaluation. See the generated scorecards for counts and uncertainty.
- Madrid: hashed before the race, verifiable after the event. The frozen forecast will be checked when race data exists.